

D605

Optimization

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A furniture maker with 45 workers decides how many tables (x) and chairs (y) to build each week. Each table earns \$70 and each chair \$30, and carpentry time is limited to 400 hours. The owner's family founded the shop in 1961. What is the objective function?

- A. Minimize $70x + 30y$
- B. Carpentry hours ≤ 400
- C. x and y
- D. Maximize $70x + 30y$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A bakery maximizes profit $5x + 4y$. The corner points of its feasible region are (0, 0), (0, 30), (20, 20), and (35, 0). The bakery sells at two farmers markets. What is the maximum profit?

- A. \$0 at (0, 0)
- B. \$360
- C. \$180 at (20, 20)
- D. \$175 at (35, 0)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An LP minimizes cost $3x + 2y$. Its corner points are (0, 12), (4, 4), and (10, 0). Which point is optimal?

- A. (10, 0) at a cost of 30
- B. (0, 12) at a cost of 24
- C. (2, 2) at a cost of 10, a cheaper point inside the region
- D. (4, 4) at a cost of 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At the optimal plan for a chemical blender, all 600 mixing hours are used while 150 of 900 storage units stay empty. The plant is in a flood zone. Which constraint can have a nonzero shadow price?

- A. Mixing hours
- B. Neither, until flood risk is added as a constraint
- C. Storage units
- D. Both constraints

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A labor constraint has a shadow price of \$12 per hour. If 50 more labor hours are added, within the allowable range, how does optimal profit change?

- A. It rises by \$62.
- B. It rises by \$600.
- C. It rises by \$12.
- D. It falls by \$600.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A charter airline's planning model says to lease 3.4 aircraft, but aircraft can only be leased whole. The airline flies 18 routes. What is the correct approach?

- A. Lease 18, one per route.
- B. Re-solve with the lease count required to be an integer.
- C. Always round up to 4.
- D. Round down to 3.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A gradient-based solver on a non-convex profit function returns different 'optimal' prices when started from different initial prices. The pricing team has 5 members. Why?

- A. The team is too small to tune the solver.
- B. Solvers always find the global optimum, so the data must be wrong.
- C. The function has several local optima; try many starting points or a global method.
- D. Linear programs have many local optima.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A startup founded by 3 engineers maximizes $8x + 5y$ subject to $x - y \leq 10$ and $x, y \geq 0$. What will the solver report?

- A. Infeasible
- B. Unbounded: profit can increase without limit.
- C. A finite optimum, because every LP has one
- D. Optimal at (10, 0) with a value of 80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A solver reports that a model is infeasible. What does that mean?

- A. There are several equally good optimal solutions.
- B. The objective can improve forever.
- C. No choice of variables satisfies all the constraints at once.
- D. The problem can never be solved under any change to the constraints.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

To minimize $f(x) = (x - 4)^2$, a solver coded in R by an intern starts at $x = 10$ with a step size of 0.1. What is x after one gradient-descent step?

- A. 8.8
- B. 11.2
- C. 4
- D. 9.4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Gradient descent is run on a convex function with a step size that is much too large. What is most likely to happen?

- A. It converges very slowly.
- B. It gets stuck in a local minimum.
- C. It always converges faster.
- D. It overshoots and oscillates or diverges.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A grocery wholesaler with 900 trucks must choose which 12 of 40 candidate sites to open. Cost for any choice comes from a black-box simulation that provides no derivatives. Which method fits?

- A.** A derivative-free heuristic such as a genetic algorithm or simulated annealing
- B.** Gradient descent
- C.** Buy more trucks to lower cost
- D.** The simplex method

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A furniture maker ~~with 45 workers~~ decides how many tables (x) and chairs (y) to build each week. Each table earns \$70 and each chair \$30, and carpentry time is limited to 400 hours. ~~The owner's family founded the shop in 1961.~~ What is the objective function?

Target word: identify the objective **Decides:** The goal is profit from tables and chairs.

A	Inverted operation	Profit is maximized, not minimized.
B	Confusion set	That is a constraint.
C	Adjacent category	Those are the decision variables.
D	Correct	The quantity to optimize is total profit.

2. Answer C Concept 2

A bakery maximizes profit $5x + 4y$. The corner points of its feasible region are (0, 0), (0, 30), (20, 20), and (35, 0). ~~The bakery sells at two farmers markets.~~ What is the maximum profit?

Target word: optimal objective value **Decides:** Evaluate every corner.

A	Inverted operation	That is the minimum.
B	Camouflage	Doubles the answer for two markets, which the model does not include.
C	Correct	Corners give 0, 120, 180, and 175; 180 is largest.
D	Dropped element	Picks the largest x without checking the other corners.

3. Answer D Concept 2 · transfer

An LP minimizes cost $3x + 2y$. Its corner points are (0, 12), (4, 4), and (10, 0). Which point is optimal?

Target word: optimal corner for a minimization **Decides:** Costs are 24, 20, 30.

A	Inverted operation	The maximum, not the minimum.
B	Dropped element	Stops at the first corner without comparing.
C	Confusion set	An LP optimum is at a corner; a point cheaper than every corner cannot be feasible.
D	Correct	The lowest-cost corner.

4. Answer A Concept 3

At the optimal plan for a chemical blender, all 600 mixing hours are used while 150 of 900 storage units stay empty. ~~The plant is in a flood zone.~~ Which constraint can have a nonzero shadow price?

Target word: which constraint is binding **Decides:** Mixing hours are fully used; storage has slack.

A	Correct	A binding constraint is the only kind that can add value when relaxed.
B	Camouflage	The flood zone does not change which constraint binds.
C	Inverted operation	Storage has slack, so extra storage is worth \$0.
D	Dropped element	Ignores the 150 unused storage units.

5. Answer B Concept 3 · transfer

A labor constraint has a shadow price of \$12 per hour. If 50 more labor hours are added, within the allowable range, how does optimal profit change?

Target word: use the shadow price **Decides:** Change = shadow price × added units.

A	Wrong stem number	Adds the two stem numbers instead of multiplying.
B	Correct	$\$12 \times 50 = \600 .
C	Dropped element	That is the value of one hour.
D	Inverted operation	Adding a binding resource raises profit.

6. Answer B Concept 4

A charter airline's planning model says to lease 3.4 aircraft, but aircraft can only be leased whole. ~~The airline flies 18 routes.~~ What is the correct approach?

Target word: handle whole-number decisions **Decides:** The decision must be an integer.

A	Camouflage	The route count is not the decision variable.
B	Correct	Integer programming finds the best feasible whole-number plan.
C	Absolute language	May be feasible but is not guaranteed optimal.
D	Dropped element	May break a capacity constraint.

7. Answer C Concept 5

A gradient-based solver on a non-convex profit function returns different 'optimal' prices when started from different initial prices. ~~The pricing team has 5 members.~~ Why?

Target word: why results vary **Decides:** The function is non-convex.

A	Camouflage	Team size does not create local optima.
B	Absolute language	Only convex problems guarantee that.
C	Correct	Local methods stop at the nearest peak.
D	Confusion set	LPs are convex; any local optimum is global.

8. Answer B Concept 6

A startup ~~founded by 3 engineers~~ maximizes $8x + 5y$ subject to $x - y \leq 10$ and $x, y \geq 0$. What will the solver report?

Target word: solution status **Decides:** y and x can both grow while $x - y$ stays ≤ 10 .

A	Confusion set	$(0, 0)$ satisfies every constraint, so it is feasible.
B	Correct	Let y grow and set $x = y + 10$; the objective grows forever.
C	Absolute language	LPs can be unbounded or infeasible.
D	Dropped element	Ignores that increasing y keeps raising the objective.

9. Answer C Concept 6 · transfer

A solver reports that a model is infeasible. What does that mean?

Target word: meaning of infeasible **Decides:** The constraints conflict.

A	Adjacent category	That is multiple optima, a feasible outcome.
B	Inverted operation	That is unbounded.
C	Correct	The feasible region is empty.
D	Absolute language	Relaxing a conflicting constraint can restore feasibility.

10. Answer A Concept 7

To minimize $f(x) = (x - 4)^2$, a solver ~~coded in R by an intern~~ starts at $x = 10$ with a step size of 0.1. What is x after one gradient-descent step?

Target word: one gradient step **Decides:** $f'(10) = 2(10 - 4) = 12$.

A	Correct	$10 - 0.1 \times 12 = 8.8$.
B	Inverted operation	Adds the gradient, moving uphill.
C	Adjacent category	The minimum itself, not where one step lands.
D	Dropped element	Uses $(x - 4) = 6$ and drops the factor of 2.

11. Answer D Concept 7 · transfer

Gradient descent is run on a convex function with a step size that is much too large. What is most likely to happen?

Target word: effect of a large step **Decides:** Steps overshoot the minimum.

A	Inverted operation	That is the too-small step problem.
B	Confusion set	A convex function has no separate local minima.
C	Absolute language	Past a point, larger steps prevent convergence.
D	Correct	Each step jumps past the minimum.

12. Answer A Concept 8

A grocery wholesaler ~~with 900 trucks~~ must choose which 12 of 40 candidate sites to open. Cost for any choice comes from a black-box simulation that provides no derivatives. Which method fits?

Target word: method for a black-box discrete choice **Decides:** No derivatives; combinatorial decision.

A	Correct	Searches the choices using only objective values.
B	Dropped element	Needs derivatives and continuous variables.
C	Camouflage	Fleet size is not the decision.
D	Adjacent category	Requires a linear objective and constraints.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Parts of an optimization model	Decision variables are what you choose; the objective is what you maximize or minimize; constraints limit the choices.	Calling a resource limit the objective, or the variables the objective.	“Decide how many,” “earn,” “limited to.”
2	Corner-point optimality in LP	A linear program with a bounded feasible region reaches its optimum at a corner point; evaluate the objective at each corner.	Picking the corner with the largest single variable, or a point inside the region.	A list of corner points and a linear objective.
3	Binding constraints and shadow prices	A binding constraint is fully used and can have a nonzero shadow price; a constraint with slack has a shadow price of zero. Change in objective $\approx$ shadow price $\times$ change in the resource (within the allowable range).	Valuing extra units of a resource that already has slack.	Resource used vs. available at the optimum.
4	Continuous vs. discrete (integer) problems	When decisions must be whole units, solve as an integer program; rounding an LP answer can be infeasible or suboptimal.	Rounding the continuous solution and assuming it is optimal.	Fractional answers for things that come in whole units.
5	Linear vs. non-linear and local optima	Linear and convex problems have one global optimum; non-convex non-linear problems can have many local optima, so results depend on the starting point.	Assuming any solver always finds the global optimum.	Different answers from different starting points.
6	Unbounded and infeasible problems	Unbounded = the objective can improve forever within the constraints; infeasible = no point satisfies all constraints.	Mixing up the two, or assuming every LP has a finite optimum.	A maximize objective with a direction left open, or constraints that conflict.
7	Gradient-based methods	To minimize, step against the gradient: $x_{\text{new}} = x - \text{step} \times f'(x)$; too large a step overshoots or diverges.	Adding the gradient (ascent) or forgetting the derivative’s factor.	A differentiable function, a starting point, and a step size.
8	Non-gradient (derivative-free) methods	When the objective has no usable derivative (black-box, discrete, or noisy), use derivative-free heuristics such as genetic algorithms or simulated annealing; they give good but not guaranteed-optimal answers.	Applying gradient descent or LP simplex to a black-box discrete choice.	A simulation-based or combinatorial objective.

Course scope (public description): Optimization is a large class of business problems requiring the iterative algorithmic maximization or minimization of one or more variables. Students in this course will select and use a variety of optimization approaches to address various business needs. The course covers classes of optimization problems at a foundational level (continuous/discrete, linear/nonlinear, and bounded/unbounded) and the solving

of linear optimization problems in both Python and R through the use of gradient and non-gradient-based algorithms. Analytics Programming is a prerequisite.